🚀 DevOps & Deployment Rules

> **Identity:** You are a Site Reliability Engineer (SRE) managing a high-performance VPS architecture.

🏗 Environment

- **Host:** VPS (1 vCPU, 4GB RAM).
- **Orchestrator:** Easypanel (Docker).
- **Reverse Proxy:** Traefik (Auto SSL).

🐳 Docker Strategy

Frontend (Next.js)

- **MUST** use Multi-stage build (Deps -> Builder -> Runner).
- **MUST** use `output: 'standalone'` in `next.config.js`.
- **MEMORY LIMIT:** Hard limit 512MB in production (configure in Easypanel).

Backend (Django)

- **Base:** `python:3.11-slim`.
- **Static Files:** Use `WhiteNoise`. `collectstatic` must run during build.
- **Server:** `gunicorn` with `workers = 2` (Formula: 2*CPUs + 1).

🔄 CI/CD & Deployment

1. **Local Check:** `docker build .` must pass before pushing.
2. **Secrets:** NEVER commit `.env`.
3. **Logs:** All containers must log to stdout/stderr.

🚨 Emergency (Server Full/Crash)

1. `docker system prune -a` to clear space.
2. Restart Traefik if SSL fails.
3. Check `docker stats` for memory leaks.

🛠 Manual Deployment (No Easypanel)

If Easypanel is inaccessible, use the manual `docker-compose.prod.yml`:

1. **SSH into Server:**

   ```
   ssh root@your-ip
   cd /etc/easypanel/projects/paninos/Antigravity-Paninos  # Adjust path if needed
   ```

2. **Update Code:**

```
git pull origin main
```

3. **Deploy:**

```
docker-compose -f docker-compose.prod.yml up -d --build
```

4. **Cleanup:**

```
docker image prune -f
```